Single-cell RNA sequence analysis for investigating tumor heterogeneity in breast cancer cells

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1. Abstract:

In this study, single-cell RNA sequence (sc-RNA seq) is used to investigate heterogeneity in gene expression data in breast cancer samples. After downloading publicly available data from Gene Expression Omnibus (GEO), raw count matrices have been evaluated using Seurat pipeline in R. With the application of quality control, normalization, dimensionality reduction and clustering, distinct cellular subpopulations were spotted. The UMAP visualization shows some dominant and rare clusters, indicating heterogeneity in cell subpopulations. Marker gene analysis revealed cell-type specific expression pattern, especially for NADK. These findings helped in figuring tumor heterogeneity and biologically relevant cell subpopulations.

2. Introduction:

Understanding the concept of cellular heterogeneity is required in the study related to breast cancer, where the tumor microenvironment has diversity of cells. The single-cell RNA sequence technique is applied to study gene expression of individual cells, rather than using multiple number of cells. Here, scRNA sequencing is used to examine the same at single cell resolutions from diverse breast cancer specimens. In this project, raw data from the Gene Expression Omnibus dataset was used to identify different cell patterns and analysis of marker gene.

3. Objective:

The objective of the project is to apply scRNA-seq for the analysis of gene expression in breast cancer dataset to find distinct cell types along with the analysis of marker gene in figuring biological identity.

4. Dataset:

Gene Expression Omnibus(GEO) gives the raw count matrix and it was imported into R using Seurat.

- Dataset ID: GSE176078
- **Data type:** Single-cell RNA-seq
- **Input:** Raw count matrix (GSE176078_RAW.tar)

5. Methods:

5.1. Quality control:

Quality control metrices are:

- nFeature_RNA, which shows number of unique genes expressed in each cell.
- nCount_RNA (Total UMI counts per cell), which represents total RNA molecules, captured per cell.
- Percent.mito, which represents percentage of reads from mitochondrial genes.

Visualization required help of:

- Violin plots for evaluating the distribution of QC metrices
- Scatter plot between sequencing depth and mitochondrial gene expression

Cells were filtered using:

- nFeature_RNA > 200
- nFeature_RNA < 6000
- percent.mito < 10

5.2. Normalization:

Data was normalized using LogNormalize. The identification of highly variable genes has been done. Data was scaled.

5.3. Dimensionality Reduction (PCA):

Principal component analysis is performed. An elbow plot decides the number of principal components, one should keep.

5.4. Clustering analysis:

Using nearest neighbour-relationship, graph-based clustering is used.

5.5. Visualization by UMAP:

Uniform Manifold Approximation and Projection (UMAP) was used to visualize the data in 2D.

5.6. Analysis of Marker genes:

This analysis has been done using FindAllMarkers(). Here, FeaturePlot is used for NADK expression.

6. Result:

The QC plot shows predominant group of high-quality cells and moderate gene count and low level of mitochondrial expression.

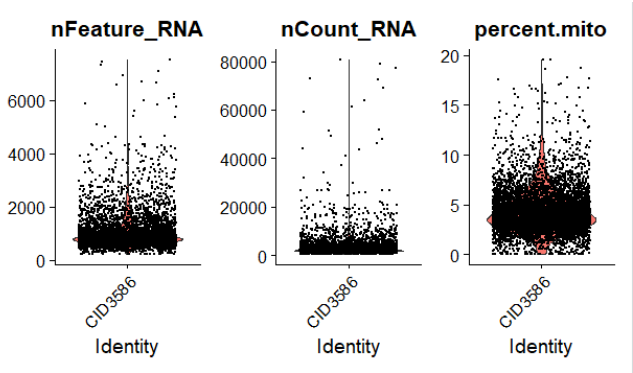

Fig 1: Violin plot

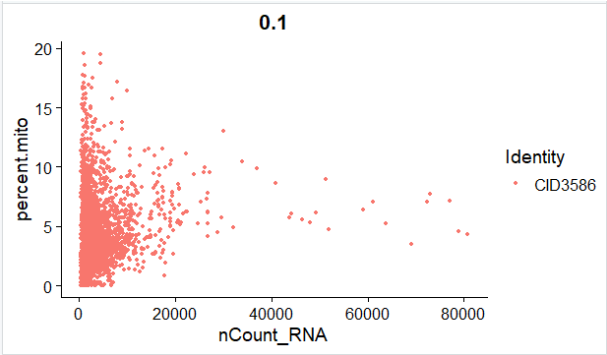

Fig 2: Scatter plot

By PCA, 5 to 6 principal components were selected for downstream evaluation.

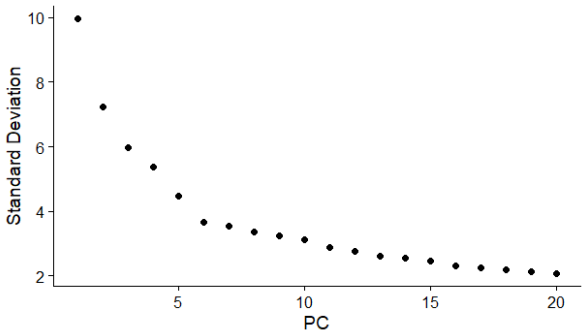

Fig 3: Elbow plot

Cells are organized into 14 clusters.

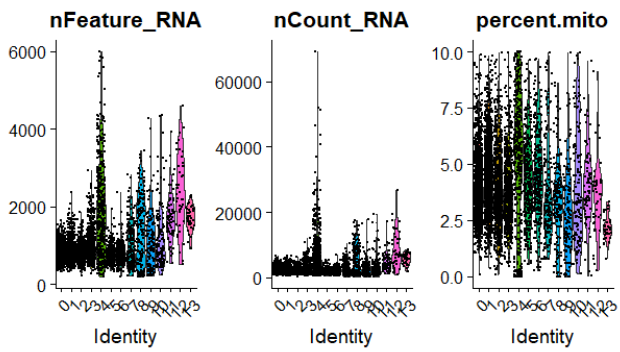

Fig 4: Clustering analysis

The UMAP visualization shows a dominant cell population and some small and separated clusters of cells.

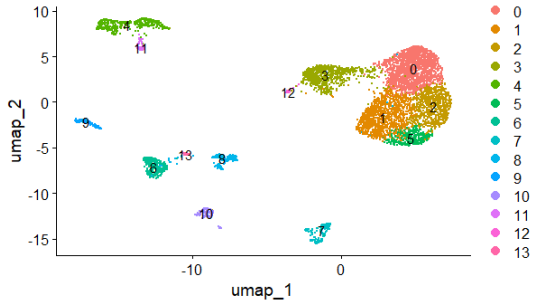

Fig 5: UMAP Visualization:

Marker genes help in finding the biological identities of group of cells.

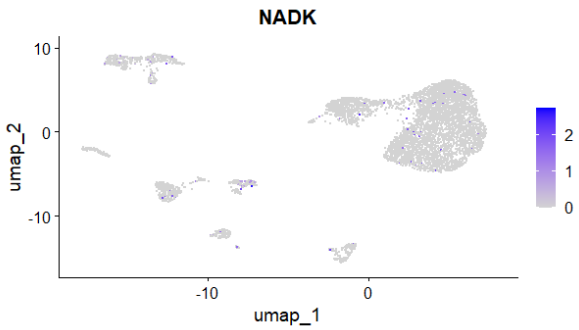

Fig 6: Marker gene analysis

7. Discussion:

The similarity between the violin plots before and after filtering indicates good data quality.

The filtering removes dead/dying cells, low-quantity cells or the doublets. Normalization adjusts expression values for following analysis.

Normalization confirms that the differences across the cells are meaningful biologically. The clustering revealing different groups of transcriptionally related cells. Cells, staying close, reveal similar kind of gene expression, while cells in a distance, have different identities. NADK is not uniformly expressed, which suggests cell-type specificity.

8. Limitations:

- Restricted biological interpretation of clusters of cells.

9. Future work:

- Further exploration of cellular states and disease conditions and treatment options can be done.
- Pathway enrichment analysis can be helpful to find biological information regulated by cluster-specific gene regulation.

10. Conclusion:

The analysis converted raw sc-RNA information into organized format. By using quality control, normalization, reduction of dimension and clustering methods, different cellular groups are found. The UMAP visualization expresses cells with similar kind of gene expression and marker genes help in deciding transcriptional heterogeneity. This interpretation provides foundation for annotation of cell-types.

11. Key words:

tumor heterogeneity, breast cancer, PCA, Clustering, UMAP, Marker gene

12. Reference:

[1] Karaayvaz, M., Cristea, S., Gillespie, S. M., Patel, A. P., Mylvaganam, R., Luo, C. C., … & Ellisen, L. W. (2018). Unravelling subclonal heterogeneity and aggressive disease states in TNBC through single-cell RNA-seq. *Nature Communications*, 9, 3588. https://doi.org/10.1038/s41467-018-06052-0

[2] Wu, S. Z., Al-Eryani, G., Roden, D. L., Junankar, S., Harvey, K., Andersson, A., … & Swarbrick, A. (2021). A single-cell and spatially resolved atlas of human breast cancers. *Nature Genetics*, 53(9), 1334–1347. https://doi.org/10.1038/s41588-021-00911-1